

Report 1

Mathematical Statistics

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1 Task 1

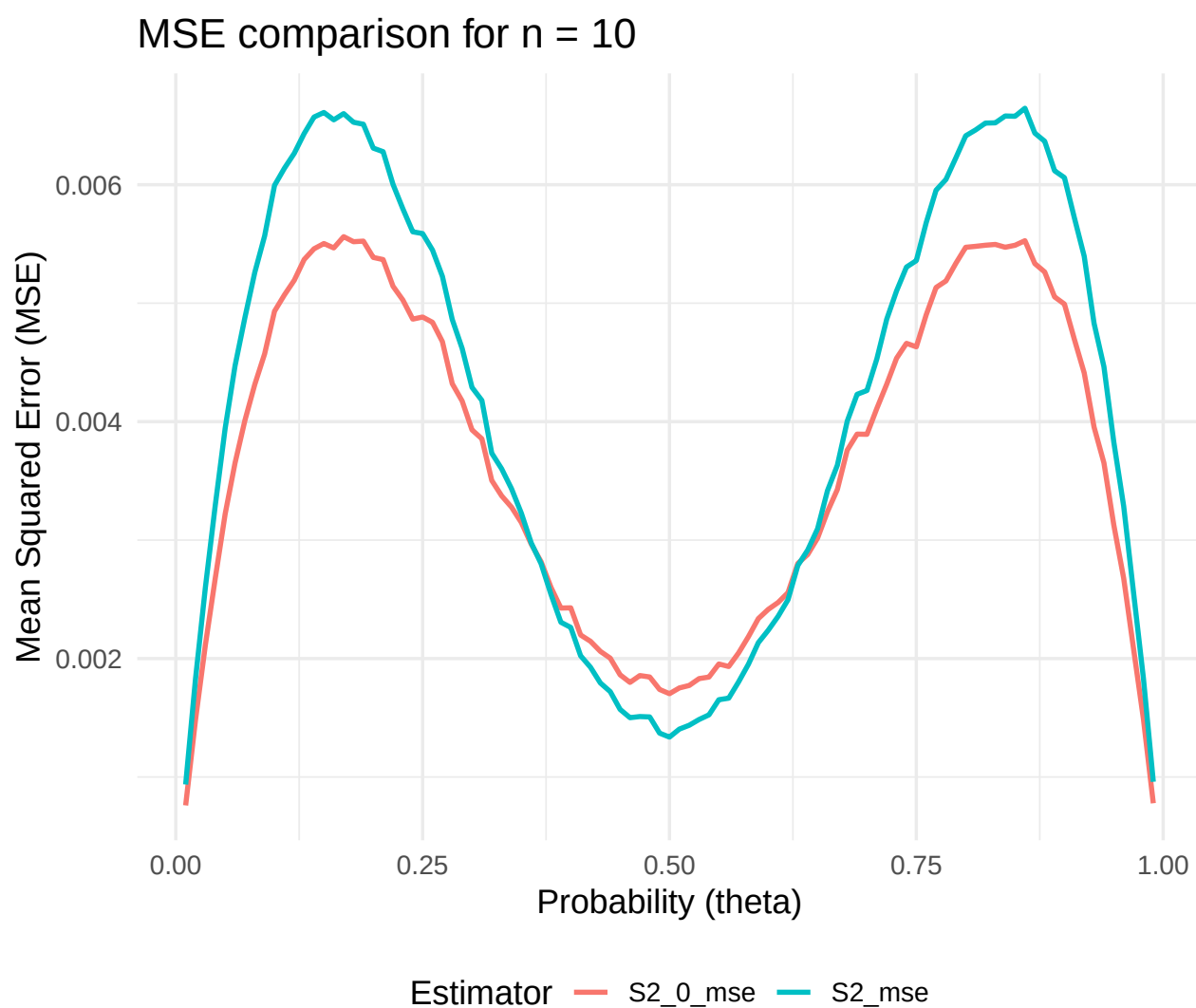


Figure 1: MSE graphs for different parameters

MSE comparison for $n = 20$

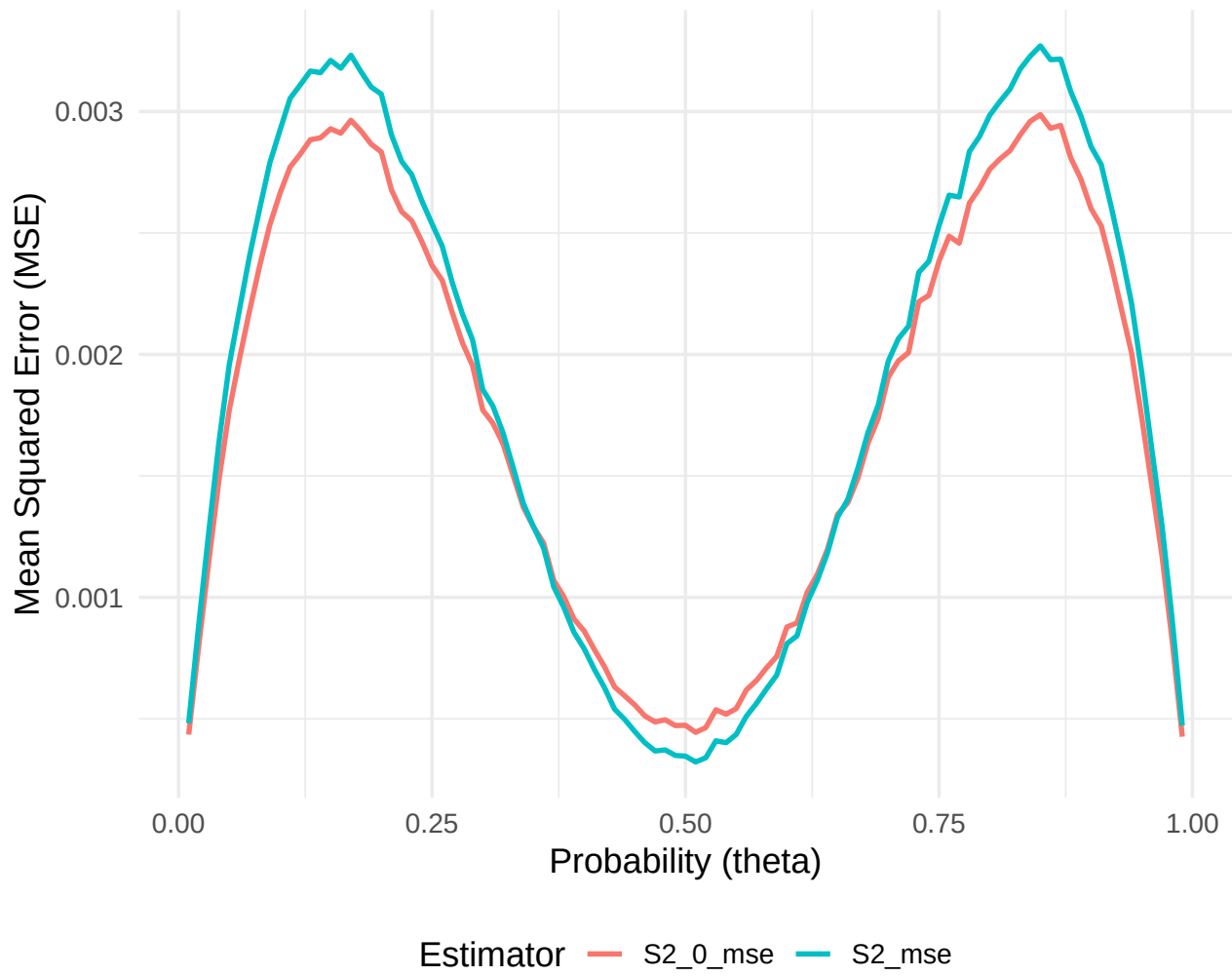


Figure 2: MSE graphs for different parameters

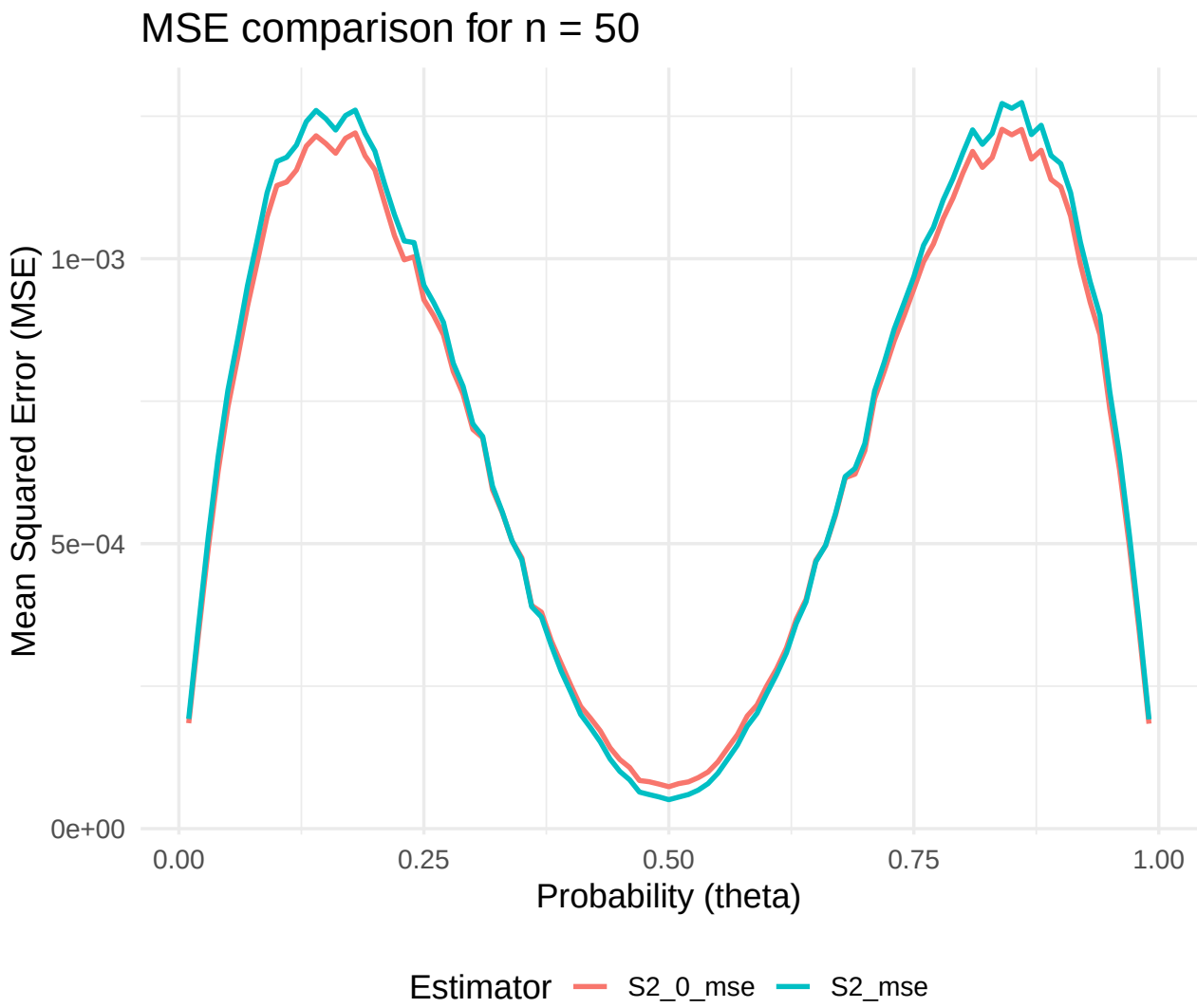


Figure 3: MSE graphs for different parameters

The first task consisted of defining a function that, for given values of n and θ , computes the estimated MSE (Mean Squared Error) for two variance estimators:

$$S^2 = \frac{1}{n-1} \sum_{k=1}^n (X_k - \bar{X})^2$$

and

$$S_0^2 = \frac{1}{n} \sum_{k=1}^n (X_k - \bar{X})^2,$$

where $X_i \sim B(n, \theta)$.

A total of 10,000 simulations were performed for $\theta \in (0, 1)$ and for several different sample sizes n . The plots present the estimated MSE values for both estimators. An important observation is that the same general pattern appears for all considered sample sizes, namely $n = 10, 20$, and 50 . When θ takes values far from $\frac{1}{2}$, the MSE is larger for the unbiased estimator S^2 . On the other hand, when θ is closer to $\frac{1}{2}$, the unbiased estimator achieves a lower MSE.

Additionally, as expected, the error decreases with increasing sample size. It can also be observed that for sufficiently large n , both estimators perform almost equally well.

In summary, the estimator S_0^2 generally performs better despite being biased. However, it also has a lower variance, which is particularly advantageous for smaller sample sizes. This estimator may therefore be preferable in situations where prior knowledge suggests that the parameter θ is likely to be close to $\frac{1}{2}$.

2 Task 2

The objective of this task is to analyze the estimation of the parameter $\theta \in (0, 1)$, representing the probability of success in n Bernoulli trials described by the binomial distribution $B(n, \theta)$. We observe the random variable X , corresponding to the number of successes.

As a measure of estimation error, we use the entropy loss function (Kullback–Leibler divergence):

$$L(\theta, d) = \theta \log \left(\frac{\theta}{d} \right) + (1 - \theta) \log \left(\frac{1 - \theta}{1 - d} \right),$$

where d denotes the estimator value. This function measures the "distance" between the distribution with parameter θ and the distribution with parameter d .

We analyze the family of estimators $d_{a,b}(X)$ defined as

$$d_{a,b}(X) = \frac{X + a}{n + a + b}.$$

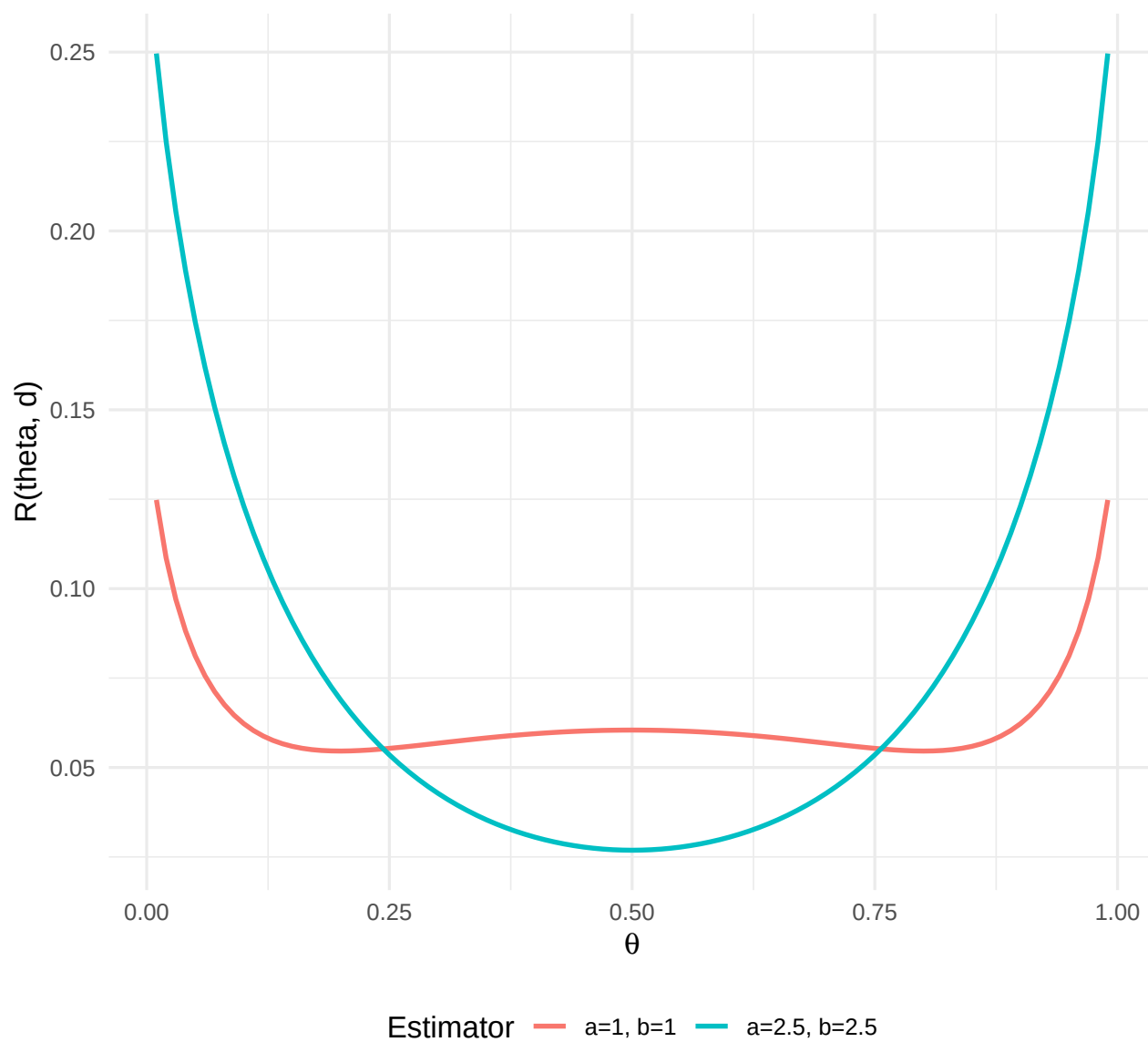
The risk function $R(\theta, d)$ is defined as the expected loss for a fixed value of θ :

$$R(\theta, d_{a,b}) = E_{\theta} [L(\theta, d_{a,b}(X))] = \sum_{x=0}^n L(\theta, d_{a,b}(x)) \cdot P(X = x \mid \theta),$$

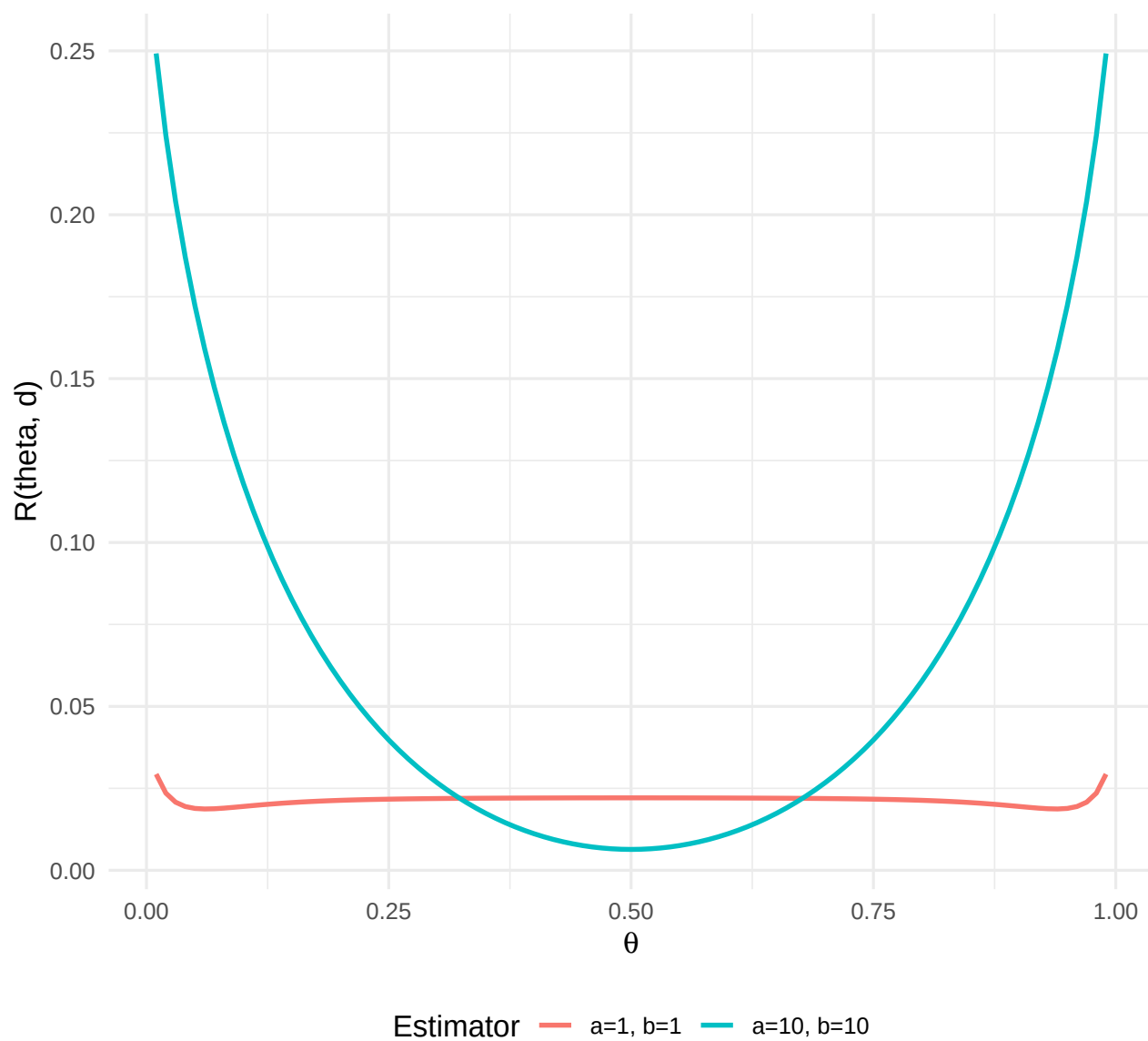
where $P(X = x \mid \theta)$ is the probability mass function of the binomial distribution $B(n, \theta)$.

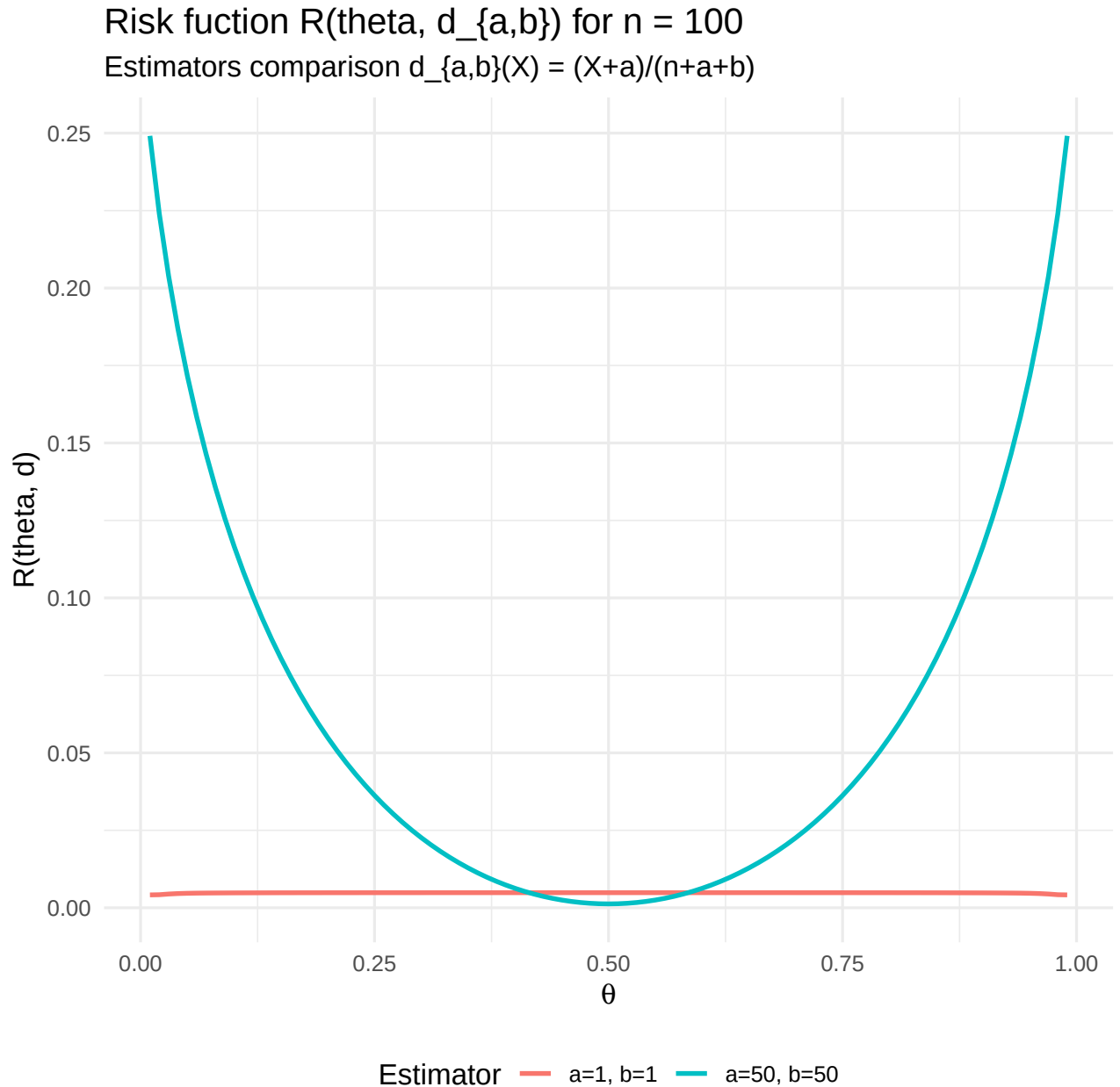
The goal of this task is to numerically compute and compare the risk functions $R(\theta, d_{a,b})$ for different sample sizes $n \in \{5, 20, 100\}$ in order to examine the influence of sample size on the risk of both estimators.

Risk fuction $R(\theta, d_{\{a,b\}})$ for $n = 5$
 Estimators comparison $d_{\{a,b\}}(X) = (X+a)/(n+a+b)$



Risk fuction $R(\theta, d_{\{a,b\}})$ for $n = 20$
 Estimators comparison $d_{\{a,b\}}(X) = (X+a)/(n+a+b)$





After generating and analyzing the risk function plots for $n = 5$, $n = 20$, and $n = 100$, the following conclusions can be drawn:

Behavior of estimator (a) ($a = 1, b = 1$):

This estimator exhibits desirable properties — its risk function remains relatively flat and low over the entire range $\theta \in (0, 1)$. The risk becomes slightly higher near the boundaries (close to 0 and 1) than in the central region. Overall, this estimator can be considered "safe" and robust, as it performs reasonably well regardless of the true value of θ .

Behavior of estimator (b) ($a = n/2, b = n/2$):

This estimator illustrates an important trade-off in Bayesian estimation:

- **When $\theta \approx 0.5$:**

The risk of this estimator is low and, importantly, lower than the risk of estimator (a). This occurs because the prior belief that θ is close to 0.5 is consistent with the true parameter value.

- **When $\theta \rightarrow 0$ or $\theta \rightarrow 1$:**

The risk of estimator (b) increases substantially, reaching values significantly higher than those of estimator (a). The prior information is sufficiently strong (especially for large n) to pull the estimator toward 0.5, even when the observed data (for example $X = 0$ or $X = n$) clearly suggest that θ is close to 0 or 1. As a consequence, the estimation error becomes large.

The choice between these estimators depends on the available prior knowledge. Estimator (a) is universal and robust, while estimator (b) represents a "high-risk, high-reward" strategy — it performs optimally when the parameter θ is indeed close to $\frac{1}{2}$, but may perform very poorly when this assumption is incorrect.

3 Task 3

In the previous task, we analyzed the risk function $R(\theta, d_{a,b})$, which measures the cost of using the estimator $d_{a,b}$ for a fixed value of the parameter θ .

In this task, we adopt a Bayesian perspective, where the parameter θ itself is treated as a random variable with a known prior distribution $\pi(\theta)$. Our prior knowledge before observing the data is described by the Beta(α, β) distribution.

We define the **expected risk** (or **Bayesian risk**) as the expected value of the classical risk with respect to the prior distribution:

$$r(\pi, d_{a,b}) = E_{\pi}[R(\theta, d_{a,b})] = \int_0^1 R(\theta, d_{a,b}) \pi(\theta) d\theta,$$

where

$$\pi(\theta) = \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha, \beta)}$$

is the density function of the Beta(α, β) distribution.

The quantity $r(\pi, d_{a,b})$ is therefore a single numerical measure describing the overall quality of the estimator $d_{a,b}$ under the prior distribution π .

The objective is to determine the values $a > 0$ and $b > 0$ that minimize the expected risk $r(\pi, d_{a,b})$ for a given prior distribution.

Methodology

1. The function `r_pi` implements the integral defining $r(\pi, d_{a,b})$, computed numerically using the `integrate` function.
2. Since deriving the analytical minimum with respect to a and b is difficult, a two-step numerical optimization procedure was applied:

- **Step 1:** The function `grid_search_ab` performs a global search over a grid of (a, b) values in order to identify an approximate location of the global minimum.
 - **Step 2:** The function `refine_opt_ab` uses the result obtained in the previous step as the starting point for the `optim` algorithm (Nelder–Mead method), which refines the parameter estimates more precisely.
3. The procedure was repeated for $n \in \{5, 20, 100\}$ in order to investigate the influence of sample size on the obtained results.

Table 1: Results of parameters optimazation a i b for Beta(1, 1)

n	a_{opt}	b_{opt}	Optimal risk $r(\pi, d_{a,b})$	Base risk $r(\pi, d_{1,1})$
5	0.99992	0.99995	0.06376467	0.06376467
20	1.00033	1.00010	0.02161756	0.02161756
50	1.00071	0.99998	0.00936801	0.00936801

The results of the numerical optimization are presented in Table 1. We can observe that the algorithm identifies the parameters (a, b) very effectively. Regardless of the sample size, the estimated values remain very close to the true optimal values.